GREAT Cell – THM Model Description

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Abstract

This report describes the coupled thermo-hydro-mechanical (THM) model used to simulate the GREAT cell experiments in OpenGeoSys. It covers the experimental apparatus, governing equations, constitutive models, geometry and boundary conditions, numerical implementation, benchmark stages, and the main assumptions and open points.

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1 Purpose and scope

The THM model reproduces the coupled thermal, hydraulic and mechanical response of rock samples tested in the GREAT cell, so that simulated stresses, strains, pore pressures and temperatures can be compared directly with the cell's sensor data. It serves three goals:

- **Verification and validation** of coupled THM codes (primarily OpenGeoSys) against a well-controlled, true-triaxial laboratory experiment.
- **Interpretation** of the measured strain field and injection pressures, including the stress redistribution around a central borehole and a fracture.
- **Benchmarking** within the international DECOVALEX framework, where several teams model the same experimental stages [1].

The model covers the intact sample (poroelastic, thermo-poroelastic) and the fractured sample (fluid flow in a single fracture under polyaxial stress).

2 Background and literature

The GREAT cell was introduced by McDermott et al. [2] as a true-triaxial cell for 200 mm samples that rotates the stress field without moving the sample and records surface strain with distributed fibre optics. First hydro-mechanical tests on resin analogues showed that fracture permeability follows the normal stress on the fracture as the load rotates.

Within DECOVALEX, the GREAT cell is one of the reference experiments of the SAFENET task, which brought fracture mechanics into DECOVALEX for the first time [1]. Kolditz et al. [3] describe the follow-up task SAFENET-2 (DECOVALEX 2027), which extends and validates the models from the laboratory to the field scale. It combines a hydro-mechanical route based on GREAT cell tests on Greywacke and Gneiss with a thermo-mechanical route (thermally induced slip and permeability change in fractured granite), and links both to the STIMTEC in-situ experiment at the Reiche Zeche mine in Freiberg. The GREAT cell has since been fitted with a heater, which makes thermal stages such as those modelled here possible.

Mollaali et al. [4] define the numerical benchmarks for the GREAT cell: a 2D plane-strain section with rubber sheath, PEE and DSS loads (load cases A, B, C, E and F), and stages from intact mechanical loading to injection and fracture propagation. They compare a discrete approach with lower-dimensional interface elements and a smeared phase-field approach in OpenGeoSys. The present THM model uses their geometry, Load A and rock parameters, and adds the heating and injection phases of the T and TH models.

3 The GREAT cell experiment

The GREAT cell (Geo-Reservoir Experimental Analogue Technology) loads a 200 mm × 200 mm cylindrical sample with a true-triaxial, rotatable stress field while fluid flows through it and fibre optics record surface strain [2, 5]. Table 1 summarises the apparatus.

The first tests used polymer and resin analogues (stiffness ∼3.85 GPa and ∼4.3 GPa) at 2–10 MPa rotating loads and fracture fluid pressures up to ∼4 MPa. Later campaigns and the benchmarks use Greywacke and Gneiss [4].

4 Governing equations

The model is quasi-static, fully saturated, single-phase Biot poro-elasticity extended by heat transport. Primary variables are displacement $\mathbf{u}$, pore pressure p and temperature T.

Table 1: GREAT cell specification [2].

Feature	Specification
Sample	Cylinder, ∼200 mm diameter × 200 mm length
Radial loading	8 opposing pairs of fluid-filled pressure-exerting elements (PEEs); each pair shares one pump
Seals between PEEs	Dynamic sealing strips (DSS)
Sleeve	2 mm Viton (fluoroelastomer) sheath between sample and PEEs/DSS
Design limits	100 MPa radial and axial stress, 40 MPa pore pressure, 100 °C
Stress rotation	Changing PEE pressures rotates the field without moving the sample
Heating	Electro-resistance bands around the cell body in an insulating jacket; thermocouples on inlet and outlet lines
Strain sensing	Distributed fibre optics (LUNA ODiSI-B) in a <1 mm groove; a value every 2.5 mm, 25 Hz
Fluid access	Ports in the top platen (centre and 50 mm radius); central borehole for injection

Momentum balance with Biot effective stress and thermal strain:

$$\nabla \cdot (\boldsymbol{\sigma}' - \alpha_B\, p\, \mathbf{I}) + \rho \mathbf{g} = \mathbf{0}, \qquad \boldsymbol{\sigma}' = \mathbb{C} : (\boldsymbol{\varepsilon} - \alpha_T (T - T_0)\, \mathbf{I}) \tag{1}$$

Fluid mass balance in the rock matrix (Darcy flow):

$$S\, \frac{\partial p}{\partial t} + \alpha_B\, \frac{\partial (\nabla \cdot \mathbf{u})}{\partial t} - \beta_{\text{eff}}\, \frac{\partial T}{\partial t} - \nabla \cdot \left(\frac{\mathbf{k}}{\mu} (\nabla p - \rho_f \mathbf{g}) \right) = Q_f \tag{2}$$

Fluid flow in a fracture (lubrication approximation, cubic law with hydraulic aperture w):

$$\frac{\partial w}{\partial t} + w\, S_f \frac{\partial p}{\partial t} - \nabla_\Gamma \cdot \left(\frac{w^3}{12\, \mu} \nabla_\Gamma p \right) = q_{\text{leak}} \tag{3}$$

Energy balance (conduction and advection, local thermal equilibrium):

$$(\rho c_p)_{\text{eff}}\, \frac{\partial T}{\partial t} + \rho_f c_{p,f}\, \mathbf{q} \cdot \nabla T - \nabla \cdot (\lambda_{\text{eff}} \nabla T) = Q_T \tag{4}$$

Here S is the storage coefficient, α_B the Biot coefficient, α_T the linear thermal expansion coefficient, β_{eff} the effective thermal expansion of the pore space, $\mathbf{k}$ the permeability tensor, μ the viscosity, $\mathbf{q}$ the Darcy flux and λ_{eff} the effective thermal conductivity. Purely mechanical and hydro-mechanical stages are special cases with T held constant.

5 Constitutive models and material parameters

Rock and sheath are linear-elastic and isotropic; fractures are represented either discretely or smeared [4] (Table 2).

Greywacke is the softer and more compressible rock; Gneiss is stiffer and tighter. Table 3 lists the values used in the THM model for Load A (`THM_model/make_THM_prj.py`); elasticity, Biot coefficient, porosity and solid density follow Table 1 of [4].

The solid conductivity and heat capacity are back-calculated from the bulk values for each porosity. Storage is not set directly; it follows from ϕ, α_B, β_f and the elasticity.

Table 2: Constitutive models.

Component	Constitutive model	Key parameters
Rock matrix (Greywacke, Gneiss)	Linear isotropic (thermo-)poro-elasticity	E, ν, ρ, ϕ, k, α_B, α_T, λ, c_p
Viton sheath	Linear elastic, impermeable	E, ν (much softer than the rock)
Fracture, discrete (LIE)	Joint law $\mathrm{d}\boldsymbol{\sigma}_f = \mathbf{K}\,\mathrm{d}\mathbf{w}$ with normal, shear and coupling stiffness k_{nn}, k_{tt}, k_{nt}	Initial aperture w_0, k_{nn}, k_{tt}
Fracture, smeared (VPF)	Phase-field damage (AT1/AT2), $\mathbf{K}_v = K_m\mathbf{I} + (1-v)\,\xi\,\frac{w^3}{12}(\mathbf{I}-\mathbf{n}\otimes\mathbf{n})$	Length scale ℓ, toughness G_c, aperture $w \approx h_e\varepsilon_1$
Fluid (water)	Constant or T-dependent ρ_f, μ	ρ_f, μ, compressibility, $c_{p,f}$, λ_f

Table 3: Material parameters of the THM model.

Parameter	Greywacke	Gneiss	Sheath
Young's modulus E (GPa)	26.85	83.9	0.1
Poisson's ratio ν (–)	0.27	0.21	0.4
Solid density ρ_s ($\mathrm{kg\,m^{-3}}$)	2650	2750	1200
Porosity ϕ (–)	0.005	0.001	as rock
Biot coefficient α_B (–)	0.8	0.6	0
Permeability k ($\mathrm{m^2}$)	1.409×10^{-14} (rock and sheath)		
Bulk thermal conductivity λ ($\mathrm{W\,m^{-1}\,K^{-1}}$)	2.8		
Bulk heat capacity ρc ($\mathrm{J\,m^{-3}\,K^{-1}}$)	2700×790		
Linear thermal expansion α_T ($\mathrm{K^{-1}}$)	8×10^{-6} (solid)		
Water: $\rho_f = 1000$ $\mathrm{kg\,m^{-3}}$, $\beta_f = 4.4 \times 10^{-10}$ $\mathrm{Pa^{-1}}$, $\mu = 10^{-3}$ Pa s, $c_f = 4180$ $\mathrm{J\,kg^{-1}\,K^{-1}}$, $\lambda_f = 0.6$ $\mathrm{W\,m^{-1}\,K^{-1}}$, no thermal expansion			

6 Geometry, mesh and boundary conditions

6.1 Geometry

The model is a 2D plane-strain horizontal mid-section of the sample (Fig. 1): a rock disc of radius 0.097 m (MaterialID 0) inside a 3 mm rubber sheath up to $r = 0.100$ m (MaterialID 1). The outer boundary is split into 16 PEE arcs and 16 DSS arcs. PEE 1 is centred at 0°, PEE i at $-(i-1)\times 22.5°$ (clockwise), and PEE ia lies opposite at +180°. Each PEE arc spans 18° and each DSS arc 4.5°; these widths are estimated from Fig. 4b of [4]. A probe circle at $r = 0.065$ m records volumetric strain, and the centre node carries the injection.

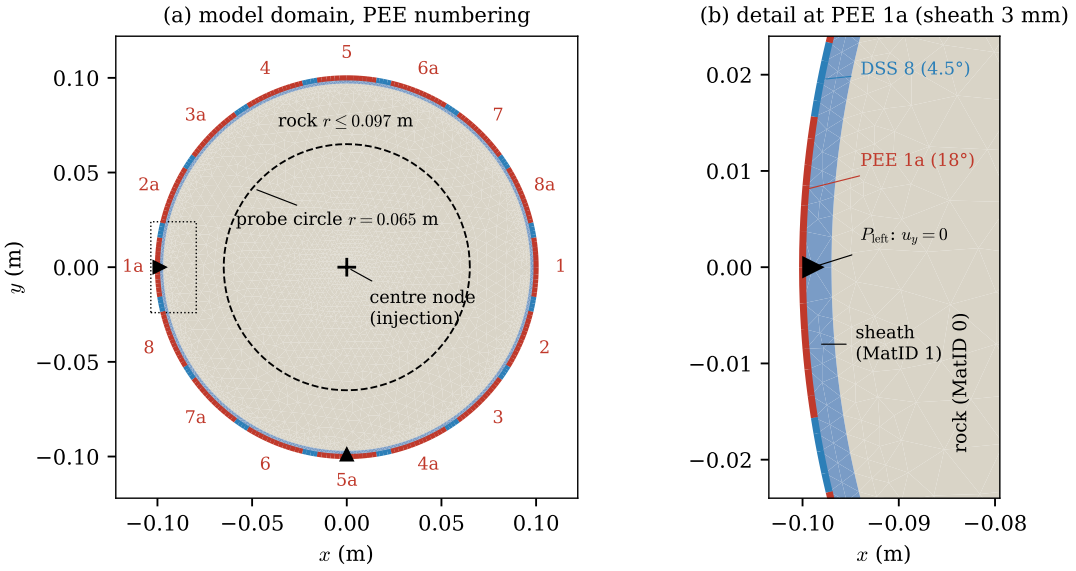


Figure 1: Model geometry. (a) Rock disc, sheath, PEE arcs (red, numbered) and DSS arcs (blue), probe circle (dashed), centre node and rigid-body points (triangles). (b) Detail of the rock–sheath boundary at PEE 1a.

6.2 Mesh

The mesh is generated with Gmsh (`make_great_cell_mesh.py`) and has 11 889 nodes and 5 804 quadratic triangles (Fig. 2). Element size is 1.5 mm at the centre, 6 mm at the probe circle and 2 mm in the sheath. Displacement uses quadratic, pressure and temperature linear shape functions (Taylor–Hood).

6.3 Boundary and initial conditions

Table 4 and Fig. 3 summarise the conditions. Load A (Table 4 of [4]) is applied as normal traction in the first time step; each DSS carries the mean of its two neighbouring PEEs, $\sigma_{\text{DSS},i} = (\sigma_{\text{PEE},i} + \sigma_{\text{PEE},i+1})/2$.

Time steps are 5 s (0–400 s), 40 s (to 4000 s), 1 s during injection, 2 s (to 4150 s) and 10 s (to 5000 s).

7 Numerical implementation in OpenGeoSys

All stages run in OpenGeoSys 6 (OGS) with the finite element method, monolithic Newton–Raphson coupling and implicit Euler time stepping (Table 5).

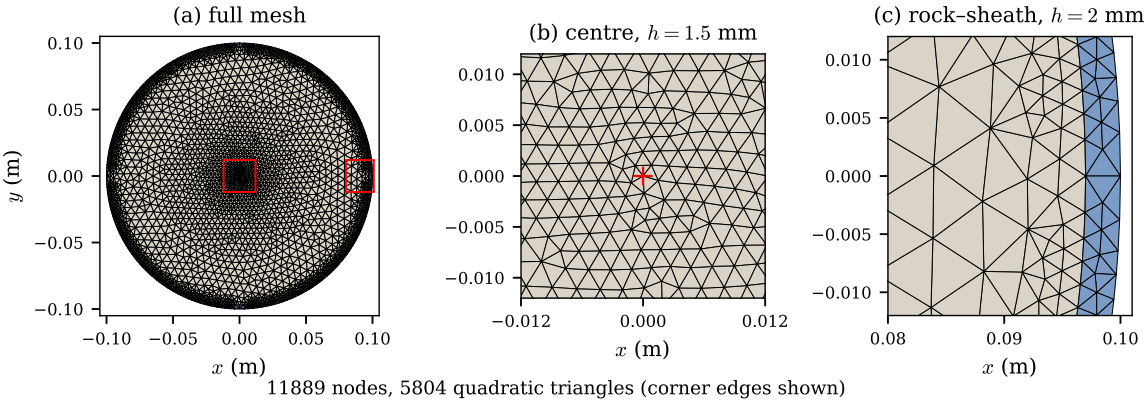


Figure 2: Finite element mesh. (a) Full mesh with the two detail windows (red). (b) Refinement around the injection node. (c) Rock–sheath transition.

Table 4: Boundary and initial conditions of the THM model, Load A.

Item	Setting
PEE tractions (MPa)	PEE 1–8 (and 1a–8a): 10.0, 6.64, 4.46, 1.17, 1.0, 3.82, 7.80, 9.95
DSS tractions	Mean of neighbouring PEEs
Rigid-body control	$P_{\text{bottom}} = (0, -0.09894)$: $u_x = u_y = 0$; $P_{\text{left}} = (-0.09894, 0)$: $u_y = 0$
Out-of-plane	Plane strain
Temperature	Outer boundary: 293 K $\rightarrow$ 363 K linearly over 0–325 s, then held
Pressure	Outer boundary drained, $p = p_0 = 1$ MPa
Injection	Centre node, 4000–4050 s: 1 l/min $= 1.667 \times 10^{-5}$ m^3 s^{-1} per m (mass rate 1.667×10^{-2} kg s^{-1} per m), $T_{\text{inj}} = 293$ K
Shut-in	4050–5000 s
Initial state	$T_0 = 293$ K, $p_0 = 1$ MPa, $\mathbf{u}_0 = \mathbf{0}$; initial effective stress $\alpha_B p_0$ so that total stress is zero

Table 5: OGS processes per stage type.

Stage type	OGS process	Fracture treatment
Mechanics only	`SMALL_DEFORMATION`	None or LIE
Hydro-mechanics, intact	`HYDRO_MECHANICS`	None
Hydro-mechanics, fractured	`HYDRO_MECHANICS` with LIE	Discrete lower-dimensional interface elements
Fracture propagation	Phase-field (VPF) HM coupling	Smeared, AT1/AT2
Thermo-hydro-mechanics	`THERMO_HYDRO_MECHANICS` (or TH2M, saturated)	None or LIE

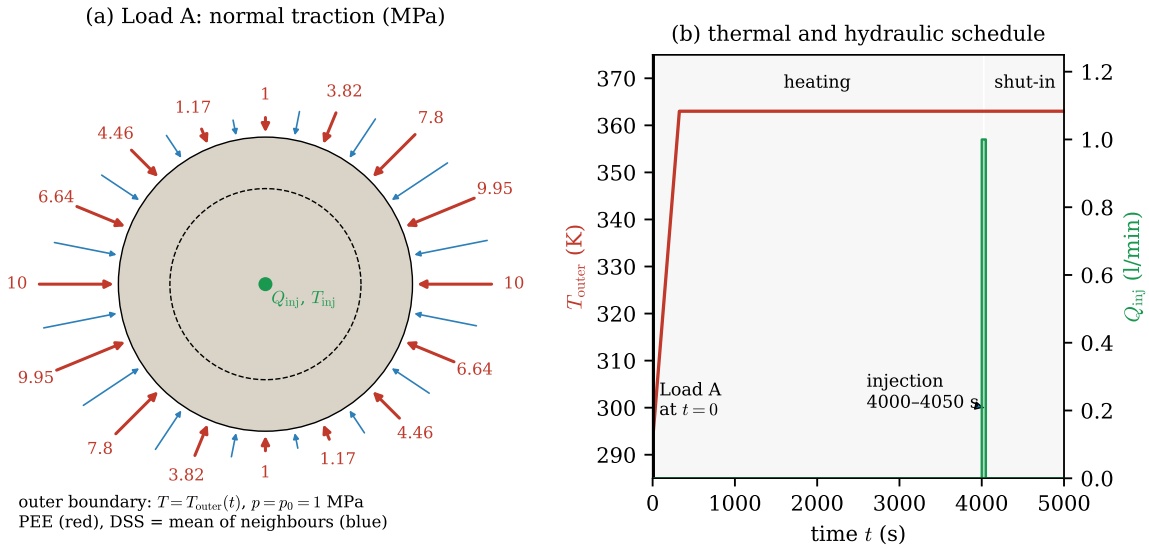


Figure 3: Boundary conditions. (a) Load A: normal traction on the PEEs (red, values in MPa) and DSSs (blue); arrow length scales with traction. (b) Outer boundary temperature and injection rate over time.

Displacements use quadratic elements and pressure and temperature linear ones (Taylor–Hood) for stable pressure fields. Pre- and post-processing (load tables per PEE, mesh generation, fibre-line extraction) run in Jupyter notebooks with `ogstools`.

The reference THM runs use OGS 6.5.9 with `THERMO_HYDRO_MECHANICS` (about 13 min per rock).

8 The OGS project

The THM model is defined by one OGS project file per rock, `THM_greywacke_loadA.prj` and `THM_gneiss_loadA.prj`, in the folder `THM_model`. Both are written by `make_THM_prj.py`, so loads and parameters are changed in the script, not in the XML. The two files differ only in the rock parameters of medium 0.

8.1 The OGS project file (PRJ)

An OGS simulation is fully defined by one project file with the extension `.prj`. It is a plain XML file with the root element `<OpenGeoSysProject>` and is passed to the solver on the command line, `ogs model.prj`. The file does not contain the mesh itself; it refers to VTU meshes stored next to it. Its main blocks are:

- `<meshes>` – the bulk mesh and the boundary and point meshes on which conditions and outputs are defined;
- `<processes>` – the physical process (here `THERMO_HYDRO_MECHANICS`), its process variables and the constitutive relations;
- `<media>` – material properties per MaterialID and phase (liquid, solid, medium);
- `<time_loop>` – time stepping, coupling, convergence criteria and output;
- `<parameters>` and `<curves>` – named constants, fields, functions and time curves used by the other blocks;

- `<process_variables>` – initial conditions, boundary conditions and source terms for each primary variable;
- `<nonlinear_solvers>` and `<linear_solvers>` – the Newton and linear solver settings.

Blocks refer to each other by name; a boundary condition, for example, names a mesh and a parameter. XML comments in the file document the model set-up.

8.2 Files and workflow

Table 6: Files of the THM model (folder `THM_model`).

File	Content
`make_great_cell_mesh.py`	Gmsh mesh and all OGS meshes (`domain.vtu`, subdomains with `bulk_node_ids`/`bulk_element_ids`); options `--lc`, `--lc_sheath`, `--lc_center`, `--order`, `--w_pee`
`make_THM_prj.py`	Writes the two project files; options `--load` (A, B, C, E, F) and `--alpha_T`
`THM_<rock>_loadA.prj`	OGS project files
`plot_THM_overview.py`	Overview plot `THM_loadA_overview.png`: centre T and p, radial profiles, $\varepsilon_{\mathrm{vol}}$ on the probe circle
`THM_out_loadA.zip`	Results and logs

The workflow is:

1. `python make_great_cell_mesh.py` – mesh and subdomain meshes;
2. `python make_THM_prj.py --load A` – project files;
3. `ogs -o out THM_greywacke_loadA.prj` (and Gneiss) – simulation, about 13 min per rock with OGS 6.5.9;
4. `python plot_THM_overview.py` – post-processing.

8.3 Structure of the project file

Table 7 maps the XML sections of the project file to the model components described above.

Two details are worth noting. First, the injection is a mass rate because the process balances mass, which is needed for the pressure-dependent water density. Second, the initial stress `sigma0` $= \alpha_B p_0$ makes the initial state ($p = p_0$, no load) free of total stress, so the response is due to Load A, heating and injection only.

9 Results

The model was built in three steps of increasing coupling: a thermal (T) model, a thermo-hydraulic (TH) model and the full THM model. Each step was checked against an analytical solution or the previous step. All runs use OGS 6.5.9. Table 8 summarises the key numbers.

Table 7: Sections of `THM_greywacke_loadA.prj`.

XML section	Content
`<meshes>`	22 meshes: `domain`, `PEE_1`–`PEE_8`, `DSS_1`–`DSS_8` (each holds arcs i and ia), `outer_boundary`, `fix_bottom`, `fix_left`, `center`, `probe_circle`
`<processes>`	`THERMO_HYDRO_MECHANICS`, integration order 4; constitutive relation `LinearElasticIsotropic` per MaterialID (0 rock, 1 sheath); `equation_balance_type = mass`; `initial_stress = sigma0`; no gravity; secondary variables $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$, Darcy velocity
`<media>`	Medium 0 (rock) and 1 (sheath), each with an `AqueousLiquid` phase (density linear in pressure, slope 4.4×10^{-10} Pa^{-1} around 1 MPa; constant μ, c, λ; no thermal expansion), a `Solid` phase (ρ_s, λ_s, c_s, α_T) and medium properties (k, ϕ, α_B, `EffectiveThermalConductivityPorosityMixing`)
`<time_loop>`	Newton with `PerComponentDeltaX` (NORM2; absolute tolerances 10^{-1} K, 10 Pa, 10^{-7} m, 10^{-7} m); `BackwardEuler`; `FixedTimeStepping` 0–5000 s in five blocks (Table 4); VTK output of `domain` at 14 fixed times, `center` and `probe_circle` at every step
`<parameters>`	Material constants, `sigma0` per MaterialID ($\alpha_B p_0$ in the rock, 0 in the sheath), initial values `T0`, `p0`, `u0`, the 16 traction values `sigma_PEE_i`/`sigma_DSS_i`, `T_outer` (`CurveScaled`), `T_inj`, and `Q_inj` (`Function`: 16.67 g s^{-1} per m for $4000 < t \leq 4050$ s)
`<curves>`	`T_outer_curve`: (0, 325, 5000) s $\rightarrow$ (293, 363, 363) K
`<process_variables>`	`temperature` (order 1): Dirichlet `T_outer` on `outer_boundary`, `DirichletWithinTimeInterval` `T_inj` on `center` (4000.5–4050 s); `pressure` (order 1): Dirichlet `p0` on `outer_boundary`, nodal source `Q_inj` on `center`; `displacement` (order 2): Dirichlet on `fix_bottom` (u_x, u_y) and `fix_left` (u_y), `NormalTraction` on the 16 PEE/DSS meshes
`<nonlinear_solvers>`	Newton, at most 20 iterations
`<linear_solvers>`	Eigen `SparseLU` with scaling

Table 8: Key results of the T, TH and THM models.

Model	Quantity	Result
T, step	Max. error vs. Bessel series	0.47 K
T, ramp	Max. error vs. Duhamel solution	0.15 K
TH	T at 4000 s vs. T model	max. difference 0.015 K
TH	Pressure vs. Thiem solution	within 0.2 %; $\Delta p = 1.09$ MPa at the centre node, 0.078 MPa at $r = 0.066$ m
TH	Cooling front at 4050 s; centre T at 5000 s	∼2 cm; 353.7 K
THM	Centre T at 4000 / 4050 / 4100 / 5000 s	356.7 / 293.0 / 309.5 / 353.6 K (both rocks)
THM	Max. Δp at the centre node	1.12 MPa (both rocks); 0.081 MPa at $r = 0.065$ m (Thiem: 0.0811)
THM	Mean $\varepsilon_{\mathrm{vol}}$ on probe circle, Greywacke	−2.46 (5 s), 11.15 (4000 s), 10.73 (4460 s), 10.94 (5000 s) $\times 10^{-2}$ %
THM	Mean $\varepsilon_{\mathrm{vol}}$ on probe circle, Gneiss	−0.95 (5 s), 12.03 (4000 s), 11.62 (4450 s), 11.82 (5000 s) $\times 10^{-2}$ %

9.1 T model

The T model (`HEAT_CONDUCTION`) solves heat conduction in a granite disc of radius 0.1 m ($\rho = 2700\,\mathrm{kg\,m^{-3}}$, $\lambda = 2.8\,\mathrm{W\,m^{-1}\,K^{-1}}$, $c_p = 790\,\mathrm{J\,kg^{-1}\,K^{-1}}$) on a mesh of 2 991 nodes. Two cases were run from $T_0 = 293$ K: a step to 323 K at the outer boundary, and the linear ramp to 363 K over 325 s used later in the THM model. OGS agrees with the analytical solutions to within 0.47 K (step) and 0.15 K (ramp), and the time discretisation converges with order ∼1, as expected for backward Euler (Fig. 4). The centre reaches the boundary temperature after about 2 h for the step case.

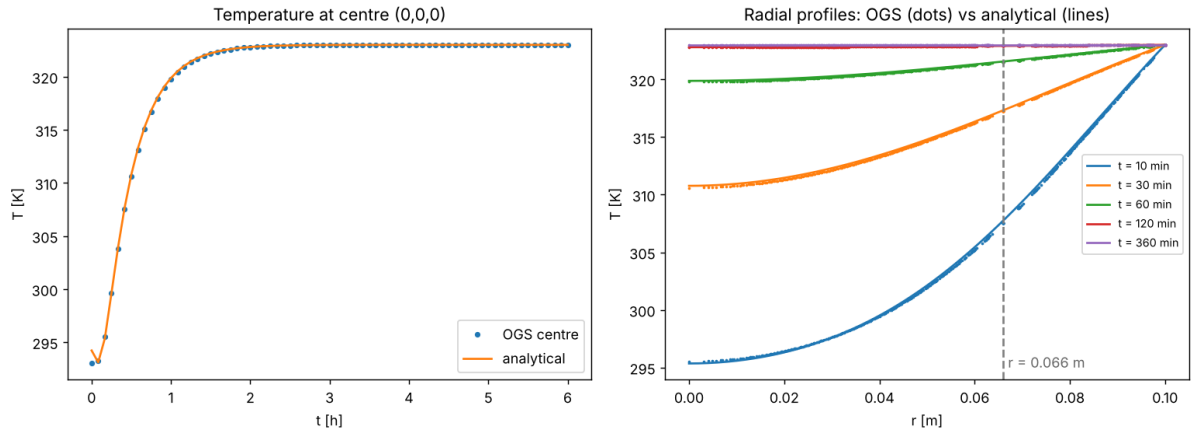


Figure 4: T model, step case: centre temperature over time (left) and radial profiles (right); OGS (dots) vs. analytical solution (lines).

9.2 TH model

The TH model (`HT`) adds Darcy flow and heat advection to the ramp case ($k = 1.409 \times 10^{-14}$ m^2, $S = 1.419 \times 10^{-11}$ Pa^{-1}, $\phi = 0.01$). Cold water (293 K) is injected at the centre at 1 l/min per

metre from 4000 to 4050 s, followed by shut-in (Fig. 5).

- **Heating phase.** Before injection the TH model reproduces the T model within 0.015 K; the centre reaches 356.7 K at 4000 s.
- **Pressure.** The pressure diffusivity $k/(\mu S) \approx 1\,\mathrm{m}^2/\mathrm{s}$ makes the pressure response quasi-instantaneous: the profile becomes the steady Thiem solution within 0.1 s and drops back to p_0 at shut-in. The centre node reaches $\Delta p = 1.09$ MPa.
- **Cooling.** Injection cools the centre to 293 K and pushes a cooling front about 2 cm into the disc by 4050 s. After shut-in, conduction from the hot boundary reheats the centre to 353.7 K at 5000 s.

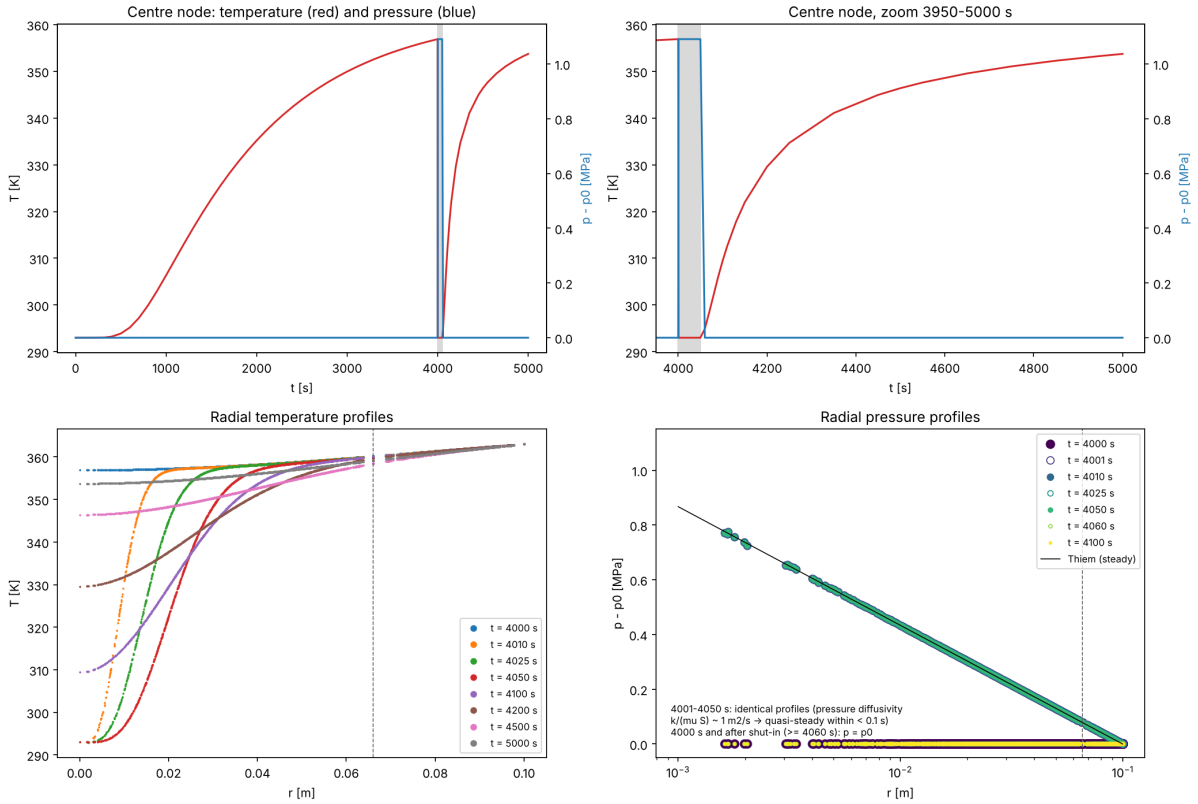


Figure 5: TH model: centre temperature and pressure (top), radial temperature profiles (bottom left) and radial pressure profiles with the Thiem solution (bottom right).

9.3 THM model

The THM model combines the TH set-up with Load A on the GREAT cell geometry for Greywacke and Gneiss (Fig. 6).

- **Temperature and pressure.** Both rocks give the same centre temperature (356.7 K at 4000 s, 353.6 K at 5000 s) and the same injection pressure (1.12 MPa at the centre node), because permeability and bulk thermal properties are equal. The centre temperature differs from the TH model by at most 0.42 K, caused by the larger time steps during heating. The pressure at $r = 0.065$ m matches the Thiem solution (0.0812 vs. 0.0811 MPa).

- **Mechanical loading.** Load A compresses the sample: mean $\varepsilon_{\mathrm{vol}}$ on the probe circle at 5 s is -2.46×10^{-2} % (Greywacke) and -0.95×10^{-2} % (Gneiss). The strain varies with angle with two maxima near 100° and 280°, and matches the M1 model to within 0.01×10^{-2} %.
- **Heating.** Thermal expansion dominates after a few hundred seconds. At 4000 s ($\Delta T \approx 67$ K on the probe circle) mean $\varepsilon_{\mathrm{vol}}$ is 11.15×10^{-2} % (Greywacke) and 12.03×10^{-2} % (Gneiss), equal to the analytical estimate $2(1+\nu)\alpha_T \Delta T + \varepsilon_{\mathrm{mech}}$. The thermal part is larger for Greywacke (13.6 vs. 13.0×10^{-2} %, higher ν), but the stiffer Gneiss carries a smaller mechanical contraction, so its net strain is higher.
- **Injection.** Cold-water injection causes a small contraction of about 0.4×10^{-2} % in both rocks, with a minimum at about 4450 s, before the strain recovers with reheating. The angular pattern of the load is preserved throughout.

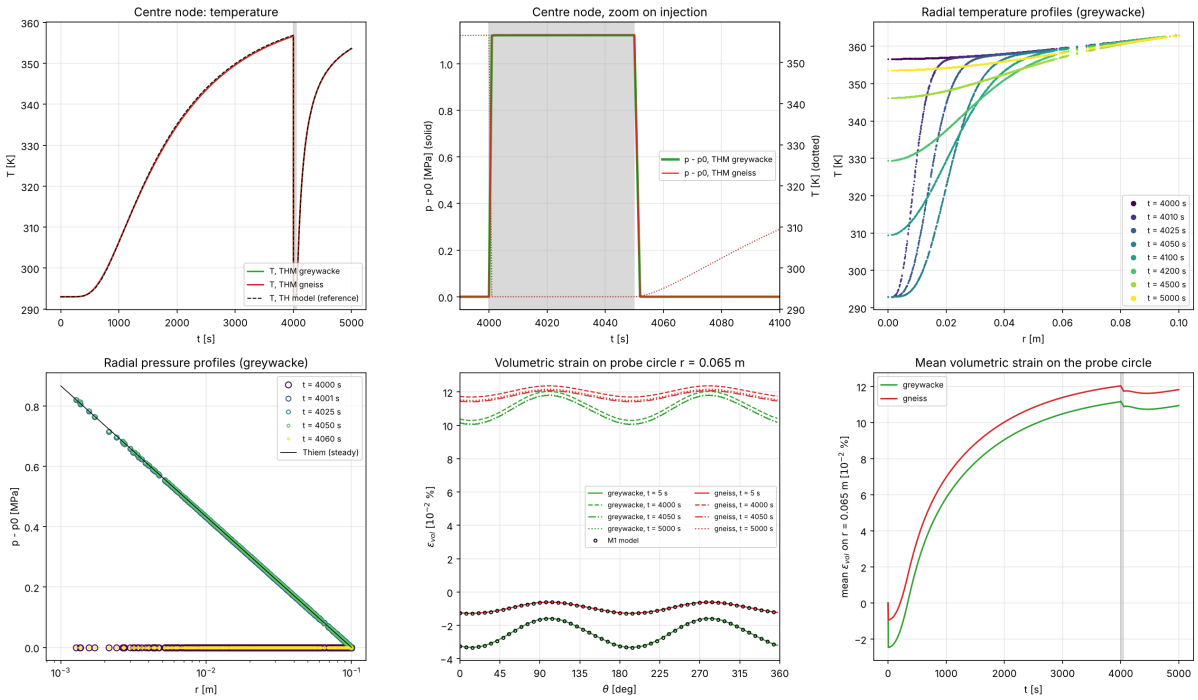


Figure 6: THM model, Load A: centre temperature compared with the TH model, centre pressure during injection, radial temperature and pressure profiles (Greywacke), volumetric strain on the probe circle vs. angle (with the M1 model at 5 s), and mean volumetric strain over time.

10 Benchmark stages and validation

The stages add one coupling at a time, so each code's error can be traced to a single process.

1. **Intact, mechanical:** Greywacke and Gneiss under Loads A–C, rotated 0°–90° in 22.5° steps. Output: circumferential strain along the fibre and volumetric strain inside $r = 0.065$ m.
2. **Intact, hydro-mechanical:** 3000 s mechanical equilibration, then 500 s borehole injection (end at 3500 s). Output: pressure and strain fields.
3. **Static fracture, M and HM:** full and half fractures, with and without inflow–outflow. Output: aperture, fracture permeability versus normal stress as the load rotates.

4. **Fracture propagation:** injection into a pre-existing fracture, and nucleation from the borehole. Output: path, breakdown pressure, opening.

5. **Thermal stages (THM):** heating of the sample by the cell bands, then cooling, under constant polyaxial load. Output: thermal strain and change of fracture permeability.

Results so far: discrete (LIE) and smeared (VPF) methods give similar volumetric strain for intact samples, and fracture permeabilities within one order of magnitude [4]. Gneiss deforms less than Greywacke; injection into Greywacke spreads strain widely, while Gneiss shows local pressure build-up. The first experiments showed that 20–30 °C heating closed the fracture plastically and lowered its permeability [2].

Validation compares simulated surface strain with the fibre-optic profiles at the mid-height of the sample, and simulated injection pressures with the pump records.

11 Assumptions, limitations and open points

- **2D plane strain** ignores axial load variation, platen friction and end effects; the fibre sits at mid-height, where this is most valid.
- **Linear elasticity** cannot capture the plastic fracture closure seen on heating, or micro-cracking near the borehole.
- **Sheath stiffness** controls how PEE pressures reach the rock; its value strongly affects simulated strain.
- **Full saturation and single-phase flow** are assumed; air in the pore space during early injection is ignored.
- **Thermal stages** are the least constrained: heat losses through platens and the frame are not yet measured.
- **Open:** the fracture-outlet back pressure is reported as 3.45 MPa in the methods and ∼3.2 MPa in the results of [2]; the benchmark uses 3.45 MPa.

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